RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE
EXPERIMENT 6B: TIME SERIES DATA ANALYSIS AND FORECAST

QUESTION:
An airline company maintains monthly records of the number of passengers travelling by air. Analyze historical airline passenger data to identify trends, seasonality, and outliers, test stationarity using the Augmented Dickey-Fuller (ADF) test, build an ARIMA forecasting model, predict future passenger numbers, and evaluate the forecasting accuracy.

AIM:
To perform time series analysis and forecasting on the Airline Passengers dataset using Python by examining trends and seasonality, testing stationarity using the ADF test, applying differencing when required, building an ARIMA model, and evaluating predictions using MAE, MSE, and RMSE.

ALGORITHM:
1. Import the required Python libraries, including Pandas, NumPy, Matplotlib, Statsmodels, and Scikit-learn.
2. Load the Airline Passengers dataset with the date column as a datetime index and display the dataset information and summary statistics.
3. Plot the original time series to observe trends, seasonality, and possible outliers.
4. Perform the Augmented Dickey-Fuller (ADF) test to determine whether the series is stationary.
5. If the series is non-stationary, apply differencing and repeat the ADF test until stationarity is achieved.
6. Split the original time series chronologically into training and testing datasets.
7. Build an ARIMA model using the selected differencing order and fit it on the training data.
8. Forecast passenger numbers for the test period and calculate MAE, MSE, and RMSE.
9. Plot the training data, testing data, and predicted passenger numbers in a single graph.
10. Interpret the forecasting results and evaluate the model's performance.

PROGRAM & OUTPUT:

```python
# Section 1: Import Libraries and Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_absolute_error, mean_squared_error

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-passengers.csv"

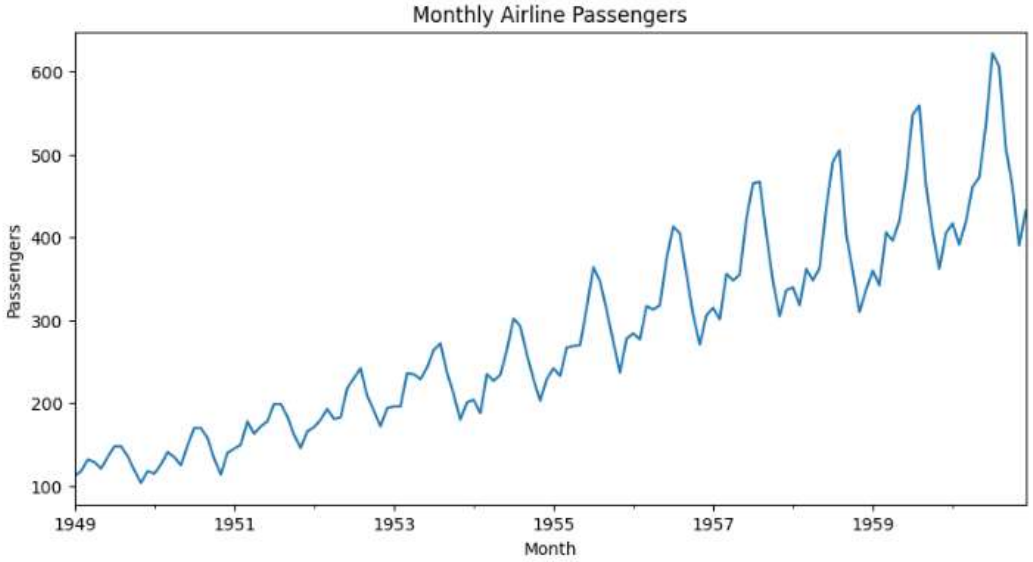
df = pd.read_csv(url, parse_dates=["Month"], index_col="Month")
df.columns = ["Passengers"]

print(df.head())
print(df.info())
print(df.describe())
```

```
            Passengers
Month
1949-01-01         112
1949-02-01         118
1949-03-01         132
1949-04-01         129
1949-05-01         121
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 144 entries, 1949-01-01 to 1960-12-01
Data columns (total 1 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Passengers  144 non-null    int64
dtypes: int64(1)
memory usage: 2.2 KB
None
       Passengers
count  144.000000
mean   280.298611
std    119.966317
min    104.000000
25%    180.000000
50%    265.500000
75%    360.500000
max    622.000000
```

```
# Name: Shyam M (241801267)
# Section 2: Plot Original Time Series

df["Passengers"].plot(figsize=(10, 5))
plt.title("Monthly Airline Passengers")
plt.xlabel("Month")
plt.ylabel("Passengers")
plt.show()
```




```
# Name: Shyam M (241801267)
# Section 3: ADF Stationarity Test

result = adfuller(df["Passengers"])

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: 0.8153688792060498
p-value: 0.991880243437641
/tmp/ipykernel_916/695558551.py:4: FutureWarning: adfuller currently returns a plain tuple whose length depends on the store and autolag arguments. In release 0.1
  result = adfuller(df["Passengers"])
```

```
# Name: Shyam M (241801267)
# Section 4: Differencing and Stationarity Check

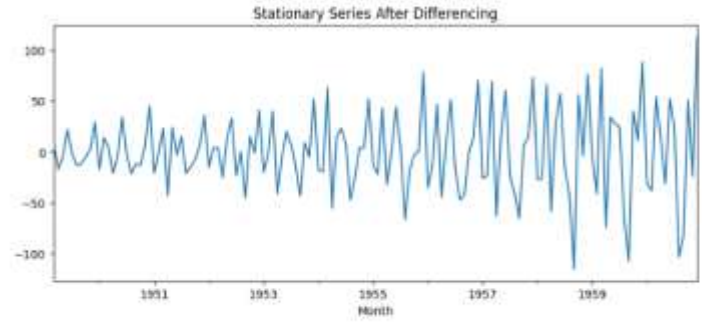
series = df["Passengers"].astype(float)
d = 0
adf_result = adfuller(series)

while adf_result[1] > 0.05 and d < 2:
    series = series.diff().dropna()
    d += 1
    adf_result = adfuller(series)

print("Differencing order (d):", d)
print("ADF p-value after differencing:", adf_result[1])

series.plot(figsize=(10, 4))
plt.title("Stationary Series After Differencing")
plt.show()
```

```
/tmp/ipykernel_916/1763003434.py:6: FutureWarning: adfuller currently returns a plain tuple whose length depends on the store and autolag arguments. In release 0.16 or
  adf_result = adfuller(series)
/tmp/ipykernel_916/1763003434.py:11: FutureWarning: adfuller currently returns a plain tuple whose length depends on the store and autolag arguments. In release 0.16 c
  adf_result = adfuller(series)
/tmp/ipykernel_916/1763003434.py:11: FutureWarning: adfuller currently returns a plain tuple whose length depends on the store and autolag arguments. In release 0.16 c
  adf_result = adfuller(series)
Differencing order (d): 1
ADF p-value after differencing: 2.7328018580143826e-29
```




```
# Name: Shyam M (241801267)
# Section 5: Split Training and Testing Data

split = int(len(df) * 0.8)

train = df["Passengers"].iloc[:split]
test = df["Passengers"].iloc[split:]

print("Training size:", len(train))
print("Testing size:", len(test))
```

```
Training size: 115
Testing size: 29
```

```
# Name: Shyam M (241801267)
# Section 6: Build and Fit ARIMA Model

model = ARIMA(train, order=(2, d, 2))
model_fit = model.fit()

print(model_fit.summary())
```

```
/usr/local/lib/python3.13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so inferred frequency MS will
  self._init_dates(dates, freq)
/usr/local/lib/python3.13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so inferred frequency MS will
  self._init_dates(dates, freq)
/usr/local/lib/python3.13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so inferred frequency MS will
  self._init_dates(dates, freq)
                               SARIMAX Results
==============================================================================
Dep. Variable:             Passengers   No. Observations:                  115
Model:                 ARIMA(2, 2, 2)   Log Likelihood                -523.084
Date:                Mon, 12 Oct 2026   AIC                           1056.169
Time:                        08:17:06   BIC                           1069.806
Sample:                  01-01-1949   HQIC                          1061.702
                       - 07-01-1958
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.5036      0.122     -4.143      0.000      -0.742      -0.265
ar.L2          0.0912      0.099      0.921      0.357      -0.103       0.285
ma.L1         -0.0552     24.247     -0.002      0.998     -47.578      47.468
ma.L2         -0.9447     22.900     -0.041      0.967     -45.828      43.939
sigma2       587.1939   1.43e+04      0.041      0.967   -2.73e+04    2.85e+04
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 2.25
Prob(Q):                              1.00   Prob(JB):                         0.32
Heteroskedasticity (H):               5.15   Skew:                             0.11
Prob(H) (two-sided):                  0.00   Kurtosis:                         2.34
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

```
# Name: Shyam M (241801267)
# Section 7: Forecast Passenger Numbers

forecast = model_fit.forecast(steps=len(test))
forecast.index = test.index

print("Forecasted Passengers:\n", forecast)
```

```
Forecasted Passengers:
 Month
1958-08-01    508.392539
1958-09-01    509.653321
1958-10-01    515.516648
1958-11-01    517.590598
1958-12-01    521.992754
1959-01-01    524.876742
1959-02-01    528.737670
1959-03-01    531.968117
1959-04-01    535.605194
1959-05-01    538.979977
1959-06-01    542.523947
1959-07-01    545.958786
1959-08-01    549.464018
1959-09-01    552.923844
1959-10-01    556.412958
1959-11-01    559.883181
1959-12-01    563.365589
1960-01-01    566.840137
1960-02-01    570.319755
1960-03-01    573.796103
1960-04-01    577.274560
1960-05-01    580.751657
1960-06-01    584.229631
1960-07-01    587.707039
1960-08-01    591.184813
1960-09-01    594.662350
1960-10-01    598.140040
1960-11-01    601.617632
1960-12-01    605.095287
Name: predicted_mean, dtype: float64
```

```
# Name: Shyam M (241801267)
# Section 8: Evaluate Forecasting Accuracy

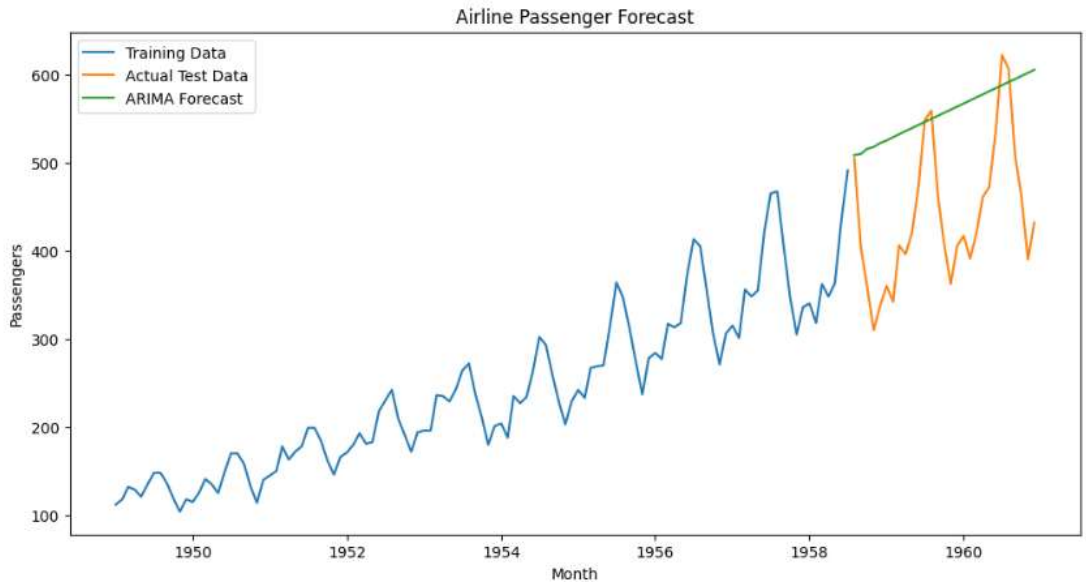
mae = mean_absolute_error(test, forecast)
mse = mean_squared_error(test, forecast)
rmse = np.sqrt(mse)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
```

```
MAE: 120.27019224229898
MSE: 18381.742370702606
RMSE: 135.5792844453112
```

```
# Name: Shyam M (241801267)
# Section 9: Plot Actual and Predicted Values

plt.figure(figsize=(12, 6))
plt.plot(train, label="Training Data")
plt.plot(test, label="Actual Test Data")
plt.plot(forecast, label="ARIMA Forecast")
plt.title("Airline Passenger Forecast")
plt.xlabel("Month")
plt.ylabel("Passengers")
plt.legend()
plt.show()
```



RESULT:

Thus, time series analysis and forecasting were successfully performed on the Airline Passengers dataset using Python. The stationarity of the time series was evaluated using the Augmented Dickey-Fuller (ADF) test, and differencing was applied when required. An ARIMA model was built to forecast passenger numbers, and its performance was evaluated using MAE, MSE, and RMSE. The actual and predicted passenger values were visualized to assess the forecasting accuracy.